

DATABRICKS AND SNOWFLAKE | CAPSTONE PROJECT

2026

Food Delivery Performance & Customer Analytics Pipeline

End-to-End Data Engineering and Business Analytics Project

Submission Detail	Information
Student Name	Chirashree Mallick
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Project Title	Food Delivery Performance & Customer Analytics Pipeline
Project Domain	Data Engineering
Primary Platform	Databricks
Target / Handover Platform	Snowflake
Project Type	End-to-End Data Engineering & Analytics Pipeline
Submission Year	2026

Prepared as an individual capstone project demonstrating Bronze → Silver → Gold data engineering, orchestration, business analytics, and Snowflake handover preparation.

Executive Summary

This project develops an end-to-end Food Delivery Performance and Customer Analytics Pipeline using Databricks and PySpark. Seven related CSV datasets covering customers, restaurants, menu items, orders, order items, delivery details, and reviews are processed through a layered Bronze, Silver, and Gold architecture.

The Silver layer addresses realistic data-quality problems including duplicate orders, orphan order-item records, invalid numeric values, missing ratings, and delivery-status edge cases. The Gold layer produces three business-ready analytical datasets: restaurant performance, customer segment performance, and category performance.

A Databricks multi-task Job automates the Bronze → Silver → Gold workflow. The Gold datasets are also exported as CSV files for Snowflake handover, with SQL prepared for stage creation, COPY INTO loading, and COPY_HISTORY verification. Actual business analysis was executed on the Gold datasets in Databricks.

1. Problem Statement

Food delivery platforms generate data across customers, restaurants, orders, menu items, deliveries, and customer reviews. When these datasets contain duplicates, invalid values, missing information, and disconnected records, direct analysis can produce inconsistent results.

The project addresses this problem by building a repeatable data pipeline that preserves raw data, applies controlled cleaning and validation, creates business-ready aggregates, and prepares analytical outputs for downstream warehouse consumption.

Item	Project Definition
Domain	Food delivery performance and customer analytics
Primary Users	Business analysts, data engineers, operations teams, and customer analytics teams
Problem	Raw operational data requires cleaning, integration, and aggregation before reliable business analysis
Success Evidence	Successful Bronze → Silver → Gold processing, successful Databricks Job run, validated Gold datasets, and three completed business analyses
Dataset	Synthetic relational food delivery dataset with 7 related CSV files and more than 50,000 records

2. Objectives and Scope

The project objectives are:

- Ingest all raw CSV files into Databricks while preserving the raw source structure in the Bronze layer.
- Clean, standardize, validate, and join related data in the Silver layer.
- Create compact Gold datasets designed around three non-trivial business questions.
- Automate the Bronze → Silver → Gold workflow using a Databricks multi-task Job.
- Export Gold datasets in CSV format for Snowflake handover and prepare Snowflake loading and analysis SQL.
- Produce measured business results from the final Gold datasets.

In Scope

- Batch processing using a Bronze/Silver/Gold architecture
- Data-quality handling and business transformations
- Databricks Job orchestration
- Gold-to-Snowflake export preparation
- Business SQL and analytical results

Out of Scope

- Real-time streaming ingestion
- Predictive machine-learning deployment
- Production CI/CD deployment
- Verified live Snowflake execution because account access was blocked during implementation

3. Solution and Features

The solution follows a medallion-style architecture. Bronze preserves source data, Silver contains cleaned and integrated entities, and Gold contains business-oriented aggregates.

- Raw data ingestion into Databricks Delta datasets.
- Duplicate removal using business keys.
- Safe numeric conversion using `try_cast` for invalid numeric values.
- Orphan order-item removal through validation against cleaned orders.
- Delivery-performance calculation including delivery delay and late-delivery indicators.
- Gold aggregation for restaurant, customer-segment, and category analysis.
- Multi-task Databricks Job with Bronze → Silver → Gold dependencies.
- CSV export of Gold datasets for Snowflake handover.

4. Tech Stack

Technology	Purpose	Use in Project
Databricks	Data engineering platform	Notebook execution, storage, workflows
PySpark / Spark SQL	Distributed processing	Cleaning, joins, transformations, aggregations
Delta Lake	Reliable analytical storage	Bronze, Silver, and Gold datasets
Databricks Workflows	Orchestration	Bronze → Silver → Gold task dependency
SQL	Analytics and warehouse scripts	Business questions and Snowflake handover
Snowflake	Target analytical warehouse	Prepared stage, COPY INTO and COPY_HISTORY scripts
VS Code	Project packaging	SQL files, README and submission structure
Git / GitHub	Version control	Final project repository and backup

5. Data Design and Layers

5.1 Source Datasets

Dataset	Raw Records	Role
customers.csv	20,000	Customer information
restaurants.csv	500	Restaurant information
menu_items.csv	5,000	Menu item and category information
orders.csv	102,000	Order transactions
order_items.csv	165,248	Items contained in orders
delivery_details.csv	100,000	Delivery timing and status information
reviews.csv	80,000	Customer ratings and reviews

5.2 Relationships

customers.customer_id → orders.customer_id
restaurants.restaurant_id → orders.restaurant_id
orders.order_id → order_items.order_id
menu_items.item_id → order_items.item_id
orders.order_id → delivery_details.order_id
orders.order_id → reviews.order_id

5.3 Layer Design

Layer	Location	Contents	Main Rules
Bronze	Databricks bronze_data	Raw source datasets	Preserve source data
Silver	Databricks silver_data	Cleaned entities + delivery performance	Type correction, deduplication, validation, joins
Gold	Databricks gold_data	3 analytical datasets	Business aggregations and metrics
Snowflake Handover	CSV export + SQL	Gold files and loading scripts	Stage/COPY INTO preparation

5.4 Gold Tables

Gold Dataset	Grain	Purpose
restaurant_performance	Restaurant	Delivery performance and customer rating analysis
customer_segment_performance	Month + customer segment	Revenue and ordering behaviour over time
category_performance	Food category	Category sales and customer rating analysis

6. System Architecture

The pipeline starts with seven raw CSV files. Databricks ingests these files into Bronze storage, transforms them into cleaned Silver datasets, and creates three business-focused Gold datasets. The Gold datasets are exported as CSV files for the planned Snowflake handover.

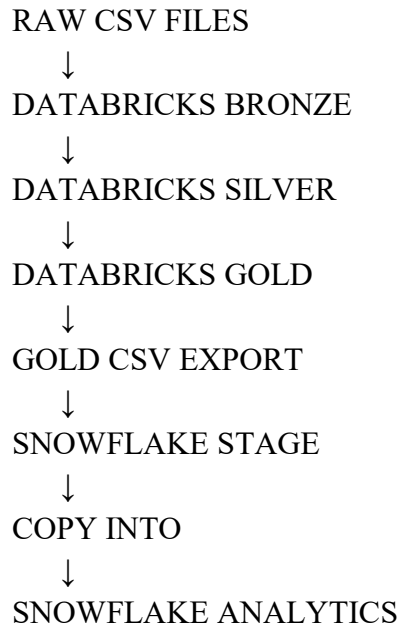


Figure 6.1: Food Delivery Analytics end-to-end architecture.

7. Implementation Highlights

7.1 Bronze Ingestion

The Bronze notebook reads the seven raw CSV datasets with headers and writes them as Delta datasets under the Bronze volume. Expected raw counts are checked before downstream processing.

```
orders_df = (
    spark.read
        .option("header", True)
        .option("inferSchema", True)
        .csv(raw_path)
)
```

7.2 Silver Data Cleaning

The Silver layer removes duplicate orders using `order_id` and safely handles invalid numeric values. Invalid values such as 'NA' are converted to NULL rather than causing the complete pipeline to fail.

```
order_items_clean = (
    order_items_df
        .withColumn("quantity", expr("try_cast(quantity AS INT)"))
        .withColumn("unit_price", expr("try_cast(unit_price AS DOUBLE)"))
)
```

```

        .withColumn("item_revenue", expr("try_cast(item_revenue AS DOUBLE)"))
    )

```

7.3 Gold Analytics

The Gold layer aggregates cleaned data into three business-oriented datasets. Restaurant performance calculates late-delivery metrics and average ratings; customer segments are aggregated by month; and categories are summarized using sales and ratings.

8. Testing and Results

Testing focused on the actual issues encountered in the pipeline and on verifying the resulting datasets.

Test / Check	Method	Observed Result	Status
Bronze source counts	Compare expected source row counts with ingestion output	All seven raw datasets ingested	PASS
Duplicate orders	Compare raw and cleaned order counts by order_id	102,000 raw → 100,000 Silver orders; 2,000 duplicates removed	PASS
Invalid order-item numerics	Use safe try_cast and inspect transformed values	Invalid numeric values handled without pipeline failure	PASS
Orphan order items	Validate order_item order_id against valid Silver orders	Orphan records removed during validation	PASS
Silver outputs	Verify Delta datasets after transformations	Cleaned entities and delivery_performance created	PASS
Gold outputs	Read three Gold Delta datasets	Restaurant, customer-segment and category datasets created	PASS
Databricks Job	Run Bronze → Silver → Gold workflow	All three tasks completed successfully	PASS
Business analysis	Execute final Gold queries	Q1, Q2 and Q3 results produced	PASS
Snowflake execution	Account/signup and live load verification	Blocked by account/network access; handover scripts prepared	BLOCKED

9. Evaluation and Metrics

9.1 Pipeline Metrics

Metric	Measured Result
Raw customers	20,000
Raw restaurants	500
Raw menu items	5,000
Raw orders	102,000
Silver orders	100,000
Duplicate orders removed	2,000
Raw order items	165,248
Raw delivery records	100,000
Raw reviews	80,000
Gold analytical datasets	3
Databricks Job tasks	3

9.2 Business Results

Question 1 – Restaurant delivery performance

After correcting the presentation of the already-percent-based `late_delivery_rate` field, the final analysis reports late-delivery percentages directly. The top restaurants in the displayed analysis were in the approximately 88.12%–90.05% range, with average customer ratings around 3.96–4.24.

Customer Segment	Total Orders	Total Revenue
Regular	39,598	63,803,902.01
Frequent	25,819	41,859,196.91
Budget	24,619	39,787,345.47
Premium	9,964	16,079,735.78

Table 9.1: Overall customer segment performance.

Category	Items Sold	Revenue	Average Rating
Desserts	38,135	18,714,029.37	4.09
Pizza	37,263	18,528,633.16	4.08
Bengali	36,624	18,296,798.61	4.07
Beverages	38,178	18,212,546.67	4.10
Chinese	37,133	18,146,177.67	4.07
Burger	36,608	17,840,841.70	4.08
Healthy	35,913	17,486,553.28	4.08
Biryani	35,426	17,481,760.14	4.10
South Indian	35,207	16,822,839.57	4.08

Table 9.2: Category performance from the Gold analysis.

10. Screenshots

The following screenshots provide evidence of the implemented pipeline. The original project folder contains the complete screenshot set; representative evidence is listed below.

Figure	Evidence File	Purpose
10.1	23052151_Bronze_01_Raw_Files.png	Bronze raw-file ingestion
10.2	23052151_Bronze_02_Row_Counts.png	Bronze row-count validation
10.3	23052151_Silver_01_Duplicate_Removal.png	Duplicate-removal evidence
10.4	23052151_Silver_04_Verification.png	Silver verification
10.5	23052151_Gold_04_Verification.png	Gold verification
10.6	23052151_Job_04_Complete_DAG.png	Databricks Job DAG
10.7	23052151_Job_05_Successful_Run.png	Successful Job execution
10.8	23052151_Snowflake_01_Gold_Export.png	Gold CSV export for handover
10.9	23052151_Final_01_Restaurant_Analysis.png	Restaurant analysis
10.10	23052151_Final_02_Customer_Analysis.png	Customer analysis
10.11	23052151_Final_03_Category_Analysis.png	Category analysis

11. Challenges and Learnings

Challenge	How It Was Solved / Handled
Duplicate orders in raw data	Orders were deduplicated by order_id; raw count 102,000 became 100,000 Silver orders.
Invalid numeric values such as NA	Safe try_cast was used for quantity, unit_price, and item_revenue.
Orphan order-item records	Order items were validated against the cleaned orders table using a valid-order join.
Column mismatch during final analysis	The Gold schema was checked and cuisine_type was used instead of the incorrect cuisine field.
Snowflake signup/network issue	Gold data export and Snowflake handover SQL were prepared; live Snowflake execution was not claimed.

Key learning: reliable analytics depends on validating each layer and documenting actual observed results rather than assuming that a successful transformation automatically means the data is correct.

12. Future Improvements

- Complete the live Snowflake stage and COPY INTO load when account access is available.
- Automate additional data-quality checks and pipeline alerts.
- Add dashboards for restaurant, customer, and category KPIs.
- Introduce incremental ingestion for newly arriving delivery data.
- Add predictive modelling for delivery-delay risk and customer retention.
- Add automated CI/CD and data-lineage controls for production deployment.

13. Conclusion

The Food Delivery Performance & Customer Analytics Pipeline demonstrates a complete data engineering workflow from raw relational CSV data to business-ready analytical outputs. Databricks was used for ingestion, cleaning, transformation, Delta storage, and orchestration.

The pipeline successfully handled duplicate orders, invalid numeric values, orphan order-item records, and other data-quality conditions. The Silver layer reduced the order dataset from 102,000 raw records to 100,000 valid orders, while the Gold layer produced three focused analytical datasets.

The resulting analysis provides measurable insights into restaurant delivery performance, customer-segment revenue and ordering behaviour, and food-category sales and ratings. The project also prepares the Gold outputs and SQL scripts for Snowflake handover while accurately documenting the live-account access limitation encountered during implementation.

14. References and Links

Reference	Details
Project Notebooks	23052151_Bronze_Ingestion.ipynb; 23052151_Silver_Transformation.ipynb; 23052151_Gold_Analytics.ipynb; 23052151_Snowflake_Handover.ipynb
SQL Files	23052151_Project_Setup.sql; 23052151_Snowflake_Handover.sql; 23052151_Business_Questions.sql
Architecture	23052151_Food_Delivery_Architecture.png
Documentation	Databricks and Snowflake course/capstone material
Source Data	Synthetic food delivery relational dataset prepared for this educational project
GitHub	To be added after final repository upload
Demo	To be added after final screen recording